Exact Coordinates for Heilbronn Configurations

$n = 5, \ldots, 9$ on the Unit Square

Companion to: N. Sudermann-Merx, *From Global Optimization to Exact Coordinates: Heilbronn's Trian*

This document lists exact coordinates of configurations for Heilbronn's triangle problem on the unit square $[0, 1]^2$ for $n = 5, \ldots, 9$. They were obtained by solving a mixed-integer nonlinear program (MINLP) with Gurobi to global optimality within solver tolerances and then deriving exact symbolic coordinates with SymPy. All triangle areas are verified in exact arithmetic. The configurations are proven optimal for $n \leq 8$ (see the paper for references). For $n = 9$, the value is a rigorous lower bound that matches the solver's numerical upper bound, but no rigorous proof of optimality is known. The derivation is fully automated by the script `exact_coordinates.py` included in this repository.

Points are labeled $1, \ldots, n$. We denote the minimum triangle area of the listed configuration by $\Delta(P)$, which equals Δ_n for $n \leq 8$.

$n = 5$: $\Delta_5 = \dfrac{\sqrt{3}}{9} \approx 0.19245$

Four critical triangles. The polynomial system of the critical triangles has a unique solution in the unit square.

Pt	x	y
1	0	$\frac{1}{3}$
2	$\frac{\sqrt{3}}{3}$	0
3	1	$1 - \frac{\sqrt{3}}{3}$
4	$\frac{2}{3}$	1
5	0	1

$n = 6$: $\Delta_6 = \dfrac{1}{8} = 0.125$

Six critical triangles. One-parameter family of optimal solutions parameterized by $c \in [0, \frac{1}{4}]$.

Pt	x	y
1	0	c
2	$\frac{1}{2}$	0
3	1	$\frac{1}{2} - c$
4	$\frac{1}{2}$	1
5	0	$c + \frac{1}{2}$
6	1	$1 - c$

$n = 7$**:** $\quad \Delta_7 = -\dfrac{a}{2a-4} \approx 0.08386$

Eight critical triangles. Coordinates are rational functions of $a \approx 0.2873$, the unique root in $(0, 1)$ of the irreducible cubic

$$a^3 + 5\,a^2 - 5\,a + 1 = 0.$$

Pt	x	y
1	0	0
2	$\dfrac{1}{a^2 - 3a + 2}$	0
3	1	a
4	$\dfrac{2a-1}{a(a-2)}$	1
5	0	1
6	$\dfrac{a-1}{a-2}$	a
7	$\dfrac{2(-a^2+3a-1)}{a^3-4a^2+5a-2}$	$-\dfrac{2a}{a-1}$

$n = 8$**:** $\quad \Delta_8 = -\dfrac{1}{36} + \dfrac{\sqrt{13}}{36} \approx 0.07238$

Twelve critical triangles. Point-symmetric configuration with the polynomial system of the critical triangles has a unique solution in the unit square, involving $\sqrt{13}$ (this does not imply global uniqueness).

Pt	x	y
1	0	0
2	$\frac{1}{6}+\frac{\sqrt{13}}{6}$	0
3	1	$\frac{7}{18}-\frac{\sqrt{13}}{18}$
4	1	1
5	0	$\frac{11}{18}+\frac{\sqrt{13}}{18}$
6	$\frac{5}{6}-\frac{\sqrt{13}}{6}$	1
7	$\frac{5}{6}-\frac{\sqrt{13}}{6}$	$\frac{7}{9}-\frac{\sqrt{13}}{9}$
8	$\frac{1}{6}+\frac{\sqrt{13}}{6}$	$\frac{2}{9}+\frac{\sqrt{13}}{9}$

$n = 9$**:** $\quad \Delta(P) = -\dfrac{11}{64} + \dfrac{9\sqrt{65}}{320} \approx 0.05488$

Eleven critical triangles. Configuration with four-fold rotational symmetry. The polynomial system of the critical triangles has a unique solution in the unit square, involving $\sqrt{65}$ (this does not imply global uniqueness).

Pt	x	y
1	0	$1-\frac{\sqrt{65}}{10}$
2	$\frac{3}{8}-\frac{\sqrt{65}}{40}$	0
3	1	$\frac{9}{16}-\frac{3\sqrt{65}}{80}$
4	$\frac{3}{8}-\frac{\sqrt{65}}{40}$	1
5	0	$\frac{5}{8}+\frac{\sqrt{65}}{40}$
6	$\frac{1}{4}+\frac{\sqrt{65}}{20}$	$\frac{3}{4}-\frac{\sqrt{65}}{20}$
7	$\frac{7}{16}+\frac{3\sqrt{65}}{80}$	0
8	$\frac{\sqrt{65}}{10}$	1
9	1	$\frac{5}{8}+\frac{\sqrt{65}}{40}$

Source: `https://github.com/spiralulam/heilbronn`